

Existing Solutions

There are already several technologies designed to help people with diabetes monitor and manage their condition. The main categories are Continuous Glucose Monitoring (CGM), diabetes-management apps, and wearable health devices.

Continuous Glucose Monitoring (CGM)

One of the most established existing solutions is Continuous Glucose Monitoring (CGM).

Examples **include FreeStyle Libre, Dexcom G7, and Medtronic Guardian**. CGM systems continuously measure glucose and provide information about glucose levels and trends.

The American Diabetes Association's 2026 Standards of Care recognizes CGM as an important part of diabetes management and recommends it for many people with diabetes depending on their treatment and individual needs.

CGM can help users:

- Monitor glucose continuously
- Identify high and low glucose
- See glucose trends
- Understand responses to food and physical activity
- Share glucose data with healthcare professionals
- Receive alerts in systems that support them

The **ADA** notes that **CGM** can help people evaluate their individual responses to treatment and lifestyle factors such as **food and physical activity**.



Diabetes Management Apps

Another existing solution is **diabetes-management software and mobile applications**.

These can help patients record and review information such as:

- Blood glucose
- Food
- Medication
- Physical activity
- Symptoms
- Other diabetes-related information

The ADA recognizes diabetes apps and other digital health technologies as part of the modern diabetes-technology landscape. It also recommends considering combinations of technologies such as **CGM, insulin pumps, and diabetes apps** with appropriate support.



Wearable Devices

Wearable technologies such as smartwatches and fitness trackers can continuously collect physiological and activity-related information.

For example:

- Heart rate
- Physical activity
- Steps
- Exercise

- Sleep
- Other physiological measurements

This is particularly interesting for our project because it provides information about **what the person is doing and how their body is responding**, rather than only measuring glucose.

Key source: American Diabetes Association, *Standards of Care in Diabetes 2026*, Section 7: Diabetes Technology. [Existing Solutions](#)

SDG 3 (Health and well-being)

Target 3.4

This is the part I would emphasize in your presentation.

Target 3.4 states:

By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being.

In simple terms, the UN wants to:

Reduce premature deaths from NCDs through prevention and treatment, while improving mental health and well-being.

And diabetes is **explicitly included** within the diseases measured under **Target 3.4**.

The UN's indicator **3.4.1** measures premature mortality from four major NCD groups:

- Cardiovascular diseases
- Cancer
- **Diabetes**
- Chronic respiratory diseases

Source: United Nations, SDG 3 / Target 3.4.

Our project be shown very simply:

SDG 3: Good Health & Well-being

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Target 3.4: Reduce premature NCD mortality

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Non-communicable diseases

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Diabetes

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Prevention + treatment + monitoring

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Your project

That gives our project a much stronger connection to SDG 3 than simply saying “*our project is related to health.*”

Diabetes affects hundreds of millions of people

According to the **World Health Organization (WHO)**, the number of people living with diabetes increased from approximately:

200 million in 1990 → 830 million in 2022.

WHO also reports that the increase has been particularly significant in **low-** and **middle-** income countries.

This demonstrates that diabetes is not a small or local problem. It is a **global public-health challenge.**

Diabetes can lead to serious health consequences

Diabetes is not simply about having high blood glucose.

Over time, it can contribute to serious complications, including:

- Blindness
- Kidney failure
- Heart attacks
- Stroke

- Lower-limb amputation

WHO reports that **diabetes and diabetes-related kidney disease caused more than 2 million deaths in 2021**. High blood glucose was also associated with approximately **11% of cardiovascular deaths**.

This directly explains why diabetes matters to **Target 3.4**.

The number of people with diabetes is expected to keep increasing

The **International Diabetes Federation (IDF)** estimates that:

589 million adults aged 20–79 were living with diabetes in 2024.

That is approximately:

1 in 9 adults worldwide.

Even more importantly, IDF estimates that approximately:

252 million adults are living with diabetes without knowing that they have it.

The IDF projects that the number could reach approximately:

853 million adults by 2050.

So the problem isn't simply:

“Many people have diabetes.”

It is:

“Diabetes is widespread, growing, and many people remain unaware of their condition.”

Another problem: Access to diabetes treatment

There is also a significant treatment gap.

WHO reported that in 2022, **more than half of people living with diabetes were not taking medication to control their diabetes**.

Treatment coverage was particularly limited in low- and middle-income countries.

This shows that:

Having diabetes does not automatically mean having sufficient support to manage it.

And diabetes management is not something that happens only when a person visits their doctor.

Sources:

United Nations — SDG 3: Good Health and Well-being

[United Nations SDG 3](#)

World Health Organization — Diabetes fact sheet

[WHO: Diabetes](#)

International Diabetes Federation — IDF Diabetes Atlas, 11th edition

[IDF Diabetes Atlas](#)
